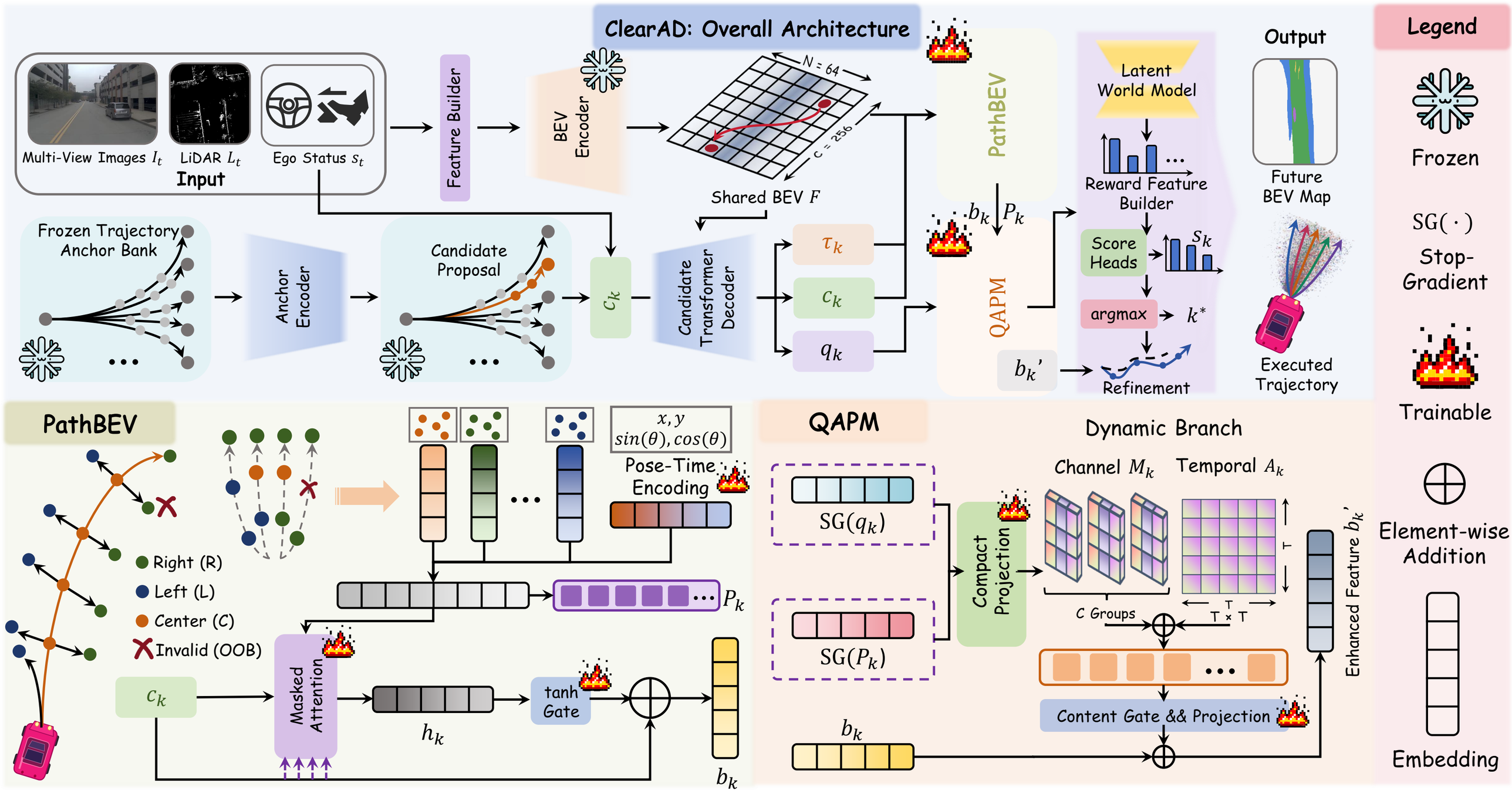

ClearAD: Overall Architecture
Input
Multi-View Images I_t
LiDAR L_t
Ego Status s_t
Feature Builder
BEV Encoder
N = 64
C = 256
Shared BEV F
Frozen Trajectory Anchor Bank
Anchor Encoder
Candidate Proposal
c_k
Candidate Transformer Decoder
τ_k
c_k
q_k
PathBEV
b_k
P_k
QAPM
b_k'
Latent World Model
Reward Feature Builder
Score Heads
S_k
argmax
k^*
Refinement
Output
Future BEV Map
Executed Trajectory
Legend
Frozen
SG(·)
Stop-Gradient
Trainable
Element-wise Addition
Embedding
PathBEV
Right (R)
Left (L)
Center (C)
Invalid (OOB)
x, y
$sin(\theta), cos(\theta)$
Pose-Time Encoding
P_k
c_k
Masked Attention
h_k
tanh Gate
b_k
QAPM
Dynamic Branch
SG(q_k)
SG(P_k)
Compact Projection
Channel M_k
Temporal A_k
C Groups
T
T × T
Content Gate && Projection
b_k
Enhanced Feature b_k'